Bach Le, Confirmed Continuing Lecturer (equivalent to U.S. Tenured Assistant Professor), Level B Step 6.
School of Computing and Information Systems, The University of Melbourne, Australia.
Australian Research Council DECRA Fellow 2022 – 2024.
Google Scholar: 2078 citations, h-index: 17, i10-index: 23.

Qualifications

- **Ph.D. Candidate | Singapore Management University, Singapore | Jan 2014 – June 2018**
  Degree conferred in June 2018.
  Thesis: "Overfitting in Automated Program Repair: Challenges and Solutions". Automatic software repair using Machine Learning, Data Mining, Program Synthesis, Analysis and Verification.
- **Ph.D. Residency Training | Carnegie Mellon University, USA | Aug 2015 – June 2016**
  Project: Automatic software repair, as part of research collaboration program SMU-CMU of Singapore Management University with Carnegie Mellon University as a strategic partner.
- **B. Tech. | Hanoi University of Science and Technology, Vietnam | Aug 2007 – May 2012**
  Degree conferred in June 2012. Degree obtained under collaboration program with Keio University, Japan.
  Thesis: HIP/SLEEK: Static verification of heap-manipulating programs via Separation Logic.

Appointments

- **Lecturer | The University of Melbourne, Australia | July 2019 – Present, Tenured from Sept 2023**
- **Fellow of Australian Research Council Discovery Early Career Researcher Award | Jan 2022 – 2024**
  *Project***:** Automated software bug fixing**, $488K as Sole PI,** 20% acceptance rate across disciplines.
- **Postdoctoral Researcher | Carnegie Mellon University, Silicon Valley, USA | June 2018 – June 2019**
  Project: ISSTAC: software vulnerabilities detection/prevention, funded by DARPA, USA.
- **Research Assistant | National University of Singapore, Singapore | Feb 2012 – Nov 2013.**
  *Project***:** HIP/SLEEK: Static verification of heap-manipulating programs via Separation Logic**.**

Teaching

Subject	Year, Level, Teaching load		#Students	Role	Performance (ESS)
Object-Oriented Software Development	Sem 1 2024	Undergraduate, 50% teaching load	~350	Coordinator	Not yet available
	Sem 1 2023			Coordinator	4.1/5.0
	Sem 1 2021			Coordinator	N/A (COVID-19)
	Sem 1 2020			Co-lecturer	N/A (COVID-19)
Programming Language Implementation	Sem 2 2020	Graduate, 50% teaching load	~40	Co-lecturer	4.4/5.0
Masters Software Engineering Project	Sem 2 2021	Graduate, 50% teaching load	~130	Co-lecturer	3.9/5.0
Models of Computation	Sem 2 2021	Undergraduate, 70% teaching load	~600	Coordinator	3.85/5.0
	Sem 2 2022				3.6/5.0
					Overall 3.97/5.0

Grants

External grants:

1. ARC DECRA 2022 (3 years, Sole PI, $488K AUD): "*Practical Automated Software Bug Fixing via Syntactic and Semantic Analyses*", $438K AUD awarded by ARC and $50K AUD awarded by The University of Melbourne. This prestigious funding from ARC has an acceptance rate of 20% across disciplines nation-wide.

Internal grants:

2. Grant for hiring a postdoc position (Level A6, ~$120K/year) starting from April 2024 to April 2025, awarded by The University of Melbourne in recognition of my DECRA 2022.
3. Seed grant, "*Formally Verified Syntax- and Semantics-guided Automatic Proof Repair*", Co-PI with Dr. Christine Rizkallah, $8K AUD, awarded by Computer Science group, The University of Melbourne.
4. Early career researcher award, "*Refixar: Practical Automated Repair of Software Regression Errors*", Sole PI, $20K AUD, awarded by Computer Science group, The University of Melbourne.

RHD Supervision and Mentoring

Postdoc @ Unimelb:

- Haoye Tian, April 2024 - now

PhD students @ Unimelb, under my primary supervision:

- Vinay Kabadi, Part-time, January 2020 – now, leave of absence on various occasions due to family issues.
  Remark: **Distinguished Paper Award** at ICSME'23.
  Second supervisor: Dr. Patanamon Thongtanunam (on long maternity leave in 2022)
- Thanh-Dat Nguyen, Full-time, April 2021 – now, leave of absence June 2021 – Jan 2022 due to COVID-19
  Remark: Plan to submit PhD thesis in October 2024.
  Second supervisor: Dr. Patanamon Thongtanunam (on long maternity leave in 2022)
  Papers during PhD: 1 x ICSME (2022, first authored), 1 x ICSE (2022, first authored), 1 x ASE (2023, second authored).
  Papers under major revision: 1 x TOSEM, 1 x ISSTA (first authored). Papers under review (submitted): 1 x TSE (first authored).
- Cong-Thanh Le, Full-time, Feb 2023 – now, second supervisor: Assoc/Prof. Toby Murray.
  Remark: **Best Undergraduate Thesis** 2019. **Google PhD Fellowship** 2023, **Amazon Applied Scientist Internship** 2024, all under my supervision on the topic of Automated Program Repair.
  Papers before PhD: 1 x ISSRE (2021, first authored), 1 x ICSE (2022, co-first authored), 1 x ICSME (2022, second authored), 1 x ESEC/FSE (2022, first authored), 1 x ICSE (2023, co-first authored).
  Papers in PhD: 1 x TSE (2023, first authored), 1 x TSE (2023, second authored), 1 x ASE (2023, third authored), 1 x TOSEM (2024, second authored), 1 x EASE (2024, second authored)
  Papers under review (submitted): 1 x TOSEM, 1 x TSE.
- Anh Ho, PhD scholarship via DECRA'22, join in June 2024, second supervisor: Dr. Christine Rizkallah.

Master's students by Research & Coursework @ Unimelb (All graduated):

- By Research: Ishan Maholtra, 2019 – 2020, Chijiang Yang, 2020-2021, Shubham Parth, 2020-2021.
- By Coursework: 40 master's students (4 groups) doing industry projects. Two groups under my primary supervision won **University Endeavor Awards in 2020 and 2021**.

Services and Leadership

International.

- **Editorial board** for Journal of Surveillance, Security, and Safety (2024 - 2026)
- Regularly serve as PC member/reviewer of various prestigious conferences and journals such as ICSE, ASE, ISSTA, TSE, TOSEM, TNSM.
- Served as chair for: Student Research Competition Track at ASE'22, Demonstration Track at IBC'22, and Software Engineering Track at SoICT'23.
- Served on an expert panel of five selected world-leading researchers on Automated Program Repair during a live event co-organized with ICSE 2023.
- Invited to Shonan Meeting (Tokyo, Japan) 2024 and Dagstuhl Seminar (Wadern, Germany) 2024 for world-leading researchers on Automated Software Debugging.

Internal. Demonstrated **commitment to growing the capacity within CIS and FEIT.**

- Hosted the President of Hanoi University of Science and Technology (HUST) in March 2024 to discuss with CIS's leadership on high-level education collaboration between CIS and HUST for a student exchange program between CIS-HUST. I am appointed by CIS's leadership to coordinate this collaboration.
- Served on DECRA's discussion panel at CIS to share experience on writing a successful DECRA application. Mentored Dr. Van-Thuan Pham (previously Lecturer at CIS) on DECRA application. Dr. Pham's has then been successfully awarded the DECRA'23 and promoted to Senior Lecturer.
- Served on postdocs hiring panel for Dr. Patanamon Thongtanunam and Dr. Van-Thuan Pham to hire postdocs to support their DECRA projects.
- Served on hiring committee of the University of Melbourne in 2022 for hiring a lecturer/senior lecturer position in CIS. Advertised the positions available and helped a colleague of mine to apply for the position.
- Served on the *Academic Advising* program to advise undergraduate students on their study and future career.

Publications (Note: My students are indicated in bold text.)

Book Chapters.

[35] *Recent Advances in Symbolic Pathfinder***.** Corina Pasareanu, Yannic Noller, Aymeric Fromherz, Bach D.X. Le, Wilem Visser, Tevfik Bultan and William Eiers. Automatic Software Verification, Book Chapter, 2023.

Fully-Referenced Conference and Journal Articles.

[34] *Does Going Beyond Branch Coverage Make Program Repair Tools More Reliable?* Amirfarhad Nilizadeh, Gary T. Leavens, Corina Pasareanu, Xuan-Bach D. Le, David Cok. The International Conference on Software Testing, Verification and Validation **(ICST)**, 2024, Research Track, 10 pages.

[33] *Refining ChatGPT-Generated Code: Characterizing and Mitigating Code Quality Issues***,** Yue Liu, **Cong-Thanh Le**, Ratnadira Widyasari, Chakkrit Tantithamthavorn, Li Li, Xuan-Bach D. Le, David Lo. Transactions on Software Engineering and Methodology **(TOSEM)**, 2024, 26 pages.

[32] *LEGION: Harnessing Pre-trained Language Models for GitHub Topic Recommendations with Distribution Balance Loss*. **Yen-Trang Dang**, **Cong-Thanh Le**, Phuc-Thanh Nguyen, Mai-Anh Bui, Phuong T. Nguyen, Bach Le, Quyet-Thang Huynh. The 28th International Conference on Evaluation and Assessment in Software Engineering **(EASE)**, 2024, Research Track, 10 pages. Note: Yen-Trang is an undergraduate female student.

[31] *Are We Ready to Embrace Generative AI for Software Q&A?* Bowen Xu, **Thanh-Dat Nguyen**, **Thanh Le-Cong**, Thong Hoang, Jiakun Liu, Kisub Kim, Chen GONG, Changan Niu, Chenyu Wang, Xuan-Bach D. Le, David Lo. The 38th IEEE/ACM International Conference on Automated Software Engineering **(ASE)**, 2023, New Ideas Track, Pages 1713-1717.

[30] *The Future Can't Help Fix The Past: Assessing Program Repair In The Wild***. (Distinguished Paper Award). Vinay Kabadi**, Dezhen Kong, Siyu Xie, G. Arthar, Lingfeng Bao, Tien-Duy BL, Xuan-Bach D. Le, and

David Lo. The 39th IEEE International Conference on Software Maintenance and Evolution **(ICSME)**, 2023, Research Track, Pages 50-61.

[29] *Multi-Granularity Detector for Vulnerability Fixes*. Truong Giang Nguyen, **Thanh Le-Cong**, Hong Jin Kang, Ratnadira Widyasari, Chengran Yang, Zhipeng Zhao, Bowen Xu, Jiayuan Zhou, Xin Xia, Ahmed Hassan, Xuan-Bach D. Le, and David Lo. Transactions on Software Engineering **(TSE)** Journal, 2023, Pages 4035-4057.

[28] *Invalidator: Automated Patch Validation via Dynamic Invariant Inference.* **Cong-Thanh Le**, Duc-Minh Luong, Xuan-Bach D. Le, David Lo, Nhat-Hoa Tran, and Quyet-Thang Huynh. Transactions on Software Engineering **(TSE)** Journal, 2023, Pages 3411-3429.

[27] *Chronos: Time-Aware Zero-Shot Identification of Libraries from Vulnerability Reports.* Y. Lyu, **Thanh Le-Cong**, Hong Jin Kang, R. Widyasari, Z. Zhao, Xuan-Bach D. Le, Ming Li, David Lo. The 45th International Conference on Software Engineering (**ICSE**), Research Track, 2023, Pages 1033-1045.

[26] *An Efficient Cyclic Entailment Procedure in a Fragment of Separation Logic*. Quang-Loc Le, Xuan-Bach D. Le**.** The 26th International Conference on Foundations of Software Science and Computation Structures (**FoSSaCS**), Research Track, 2023, Pages 477-497.

[25] *VulCurator: A Vulnerability-Fixing Commit Detector*. Truong Giang Nguyen, **Thanh Le-Cong**, Hong Jin Kang, Xuan-Bach D. Le, David Lo. The 21st Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE**), 2022, Demonstration Track, Pages 1726-1730.

[24] *AutoPruner: Tranformer-based Call Graph Pruning***. Thanh Le-Cong**, Hong Jin Kang, Truong Giang Nguyen, S. Haryono, David Lo, Xuan-Bach D. Le, and Quyet-Thang Huynh. The 21st Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE**), Research Track, 2022, Pages 520-532.

[23] *FFL: Fine grained Fault Localization for Student Programs via Syntactic and Semantic Reasoning.* **Thanh-Dat Nguyen**, **Thanh Le-Cong**, Duc-Minh Luong, Van-Hai Duong, Xuan-Bach D. Le, David Lo and Quyet-Thang Huynh. The 38th International Conference on Software Maintenance and Evolution (**ICSME**), Research Track, 2022, Pages 151-162.

[22] *Test Mimicry to Assess the Exploitability of Library*. Hong Jin Kang, Truong Giang Nguyen, Xuan-Bach D. Le, David Lo, Corina Pasareanu**.** The 31st International Symposium on Software Testing and Analysis (**ISSTA**), Research Track, 2022, Pages 276-288.

[21] *Toward the Analysis of Graph Neural Networks*. **Thanh-Dat Nguyen**, **Thanh Le-Cong**, Thanh Vu H Nguyen, Xuan-Bach D. Le, Quyet-Thang Huynh. The 44th International Conference on Software Engineering (**ICSE**), NIER Track, 2022, Pages 116-120.

[20] *REFIXAR: Multi-version Reasoning for Automated Repair of Regression Errors*. Xuan-Bach D. Le, Quang-Loc Le. The 32nd International Symposium on Software Reliability Engineering (**ISSRE**), Research Track, 2021, Pages 162-172.

[19] *Usability and Aesthetics: Better Together for Automated Repair of Web Pages*. **Cong-Thanh Le**, Xuan-Bach D. Le, Quyet-Thang Huynh, Phi Le Nguyen. The 32nd International Symposium on Software Reliability Engineering (**ISSRE**), Research Track, 2021, Pages 173-183.

[18] *More Reliable Test Suites for Dynamic Program Repair by Using Counterexamples*. Amirfarhad Nilizadeh, Marlon Calvo, Gary T. Leavens, and Xuan-Bach D. Le. The 32nd International Symposium on Software Reliability Engineering (**ISSRE**), Research Track, 2021, Pages 208-219.

[17] *Exploring True Test Overfitting in Dynamic Program Repair using Formal Methods***.** Amirfarhad Nilizadeh, Gary T. Leavens, Xuan-Bach D. Le, Corina Pasareanu, and David R. Cok. International Conference on Software Testing Verification and Validation (**ICST**), Research Track, 2021, Pages 229-240.

[16] *Smart Contract Development: Challenges and Opportunities.* Z. Weiqin, L. David, S. K. Pavneet, Xuan-Bach D. Le, F. Yang, Zhenyu Chen, and Baowen Xu. Transactions on Software Engineering ***(TSE)***, 2019, Pages 2084-2106.

[15] *SAFFRON: Adaptive Grammar-based Fuzzing for Worst-Case Analysis***.** Xuan-Bach D. Le, Corina Pasareanu, Rohan Padhye, David Lo, Willem Visser, and Koushik Sen. Java Pathfinder Workshop collocated with ASE **(JPF)**, 2019, Pages 14-18.

[14] *Symbolic Pathfinder for SV-COMP***.** Yannic Noller, Corina Pasareanu, Aymeric Fromherz, Xuan-Bach D. Le, and Willem Visser. Competition on Software Verification (**SV-COMP**), 2019, Pages 239-243.

[13] *On Reliability of Patch Correctness Assessment*. Xuan-Bach D. Le, Lingfeng Bao, David Lo, Xia Xin, Shanping Li, and Corina Pasareanu. In International Conference on Software Engineering (**ICSE**), Research Track, 2019, Pages 524-535.

[12] *Overfitting in Semantics-based Automated Program Repair***.** Xuan Bach D Le, David Lo, Ferdian Thung, and Claire Le Goues. In International Conference on Software Engineering (**ICSE**), Journal First Track, *2018*. In Empirical Software Engineering (**EMSE**) Journal, 2018, Pages 3007-3033.

[11] *S3: Syntax- and Semantics-guided Repair Synthesis via Programming by Examples***.** Xuan Bach D Le, David Lo, Duc-Hiep Chu, Claire Le Goues, and Willem Visser. In European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering (**ESEC/FSE**), Research Track, 2017, Pages 593-604.

[10] *JFix: Semantics-based Repair of Java Programs via Symbolic Pathfinder***.** Xuan Bach D Le, Duc-Hiep Chu, David Lo, Claire Le Goues, and Willem Visser. In 26th International Symposium on Software Testing and Analysis (**ISSTA**), Demo Track, 2017, Pages 376-379.

[9] *Xsearch: A Domain-specific Cross-language Relevant Question Retrieval Tool***.** Bowen Xu, Xing Zhen-chang, Xia Xin, David Lo, and Xuan-Bach D Le. In European Software Engineering Conference and the Symposium on the Foundations of Software Engineering (**ESEC/FSE**), 2017, Demo Track, Pages 1009-1013.

[8] *Towards Efficient and Effective Automatic Program Repair***.** Xuan-Bach D Le. In Proceedings of the International Conference on Automated Software Engineering (**ASE**), 2016, Doctoral Symposium, Pages 876-879.

[7] *Enhancing Automated Program Repair with Deductive Verification*. Xuan-Bach D Le, Quang Loc Le, David Lo, and Claire Le Goues. In International Conference on Software Maintenance and Evolution (**ICSME**), 2016, ERA Track, Pages 428-432.

[6] *Empirical Study on Synthesis Engines for Semantics-based Program Repair***.** Xuan Bach D Le, David Lo, and Claire Le Goues. In International Conference on Software Maintenance and Evolution (**ICSME**), 2016, ERA Track, Pages 423-427.

[5] *Recommending Code Changes for Automatic Backporting of Linux Device Drivers***.** Ferdian Thung, Xuan-Bach D Le, David Lo, and Julia Lawall. In International Conference on Software Maintenance and Evolution (**ICSME**), Research Track, 2016, Pages 222-232.

[4] *History Driven Program Repair*. Xuan Bach D Le, David Lo, and Claire Le Goues. In International Conference on Software Analysis, Evolution, and Reengineering (**SANER**), 2016, Research Track, Pages 213-224.

[3] *Should Fixing These Failures be Delegated to Automated Program Repair?* Xuan-Bach D Le, Tien-Duy B Le, and David Lo. In International Symposium on Software Reliability Engineering (**ISSRE**), Research Track, 2015, Pages 427-437.

[2] *Synergizing Specification Miners Through Model Fissions and Fusions.* Tien-Duy B Le, Xuan-Bach D Le, David Lo, and Ivan Beschastnikh. In International Conference on Automated Software Engineering (**ASE**), Research Track, 2015, Pages 115-125.

[1] *Active Semi-supervised Defect Categorization*. Ferdian Thung, Xuan-Bach D Le, and David Lo. In International Conference on Program Comprehension (**ICPC**), Research Track, 2015, Pages 60-70.